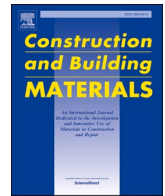




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Evaluation of high and low temperature performance characteristics of bricks fabricated with lunar regolith simulant

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ABSTRACT

For future extraterrestrial construction projects, the service performance of lunar regolith simulants under high and low temperature cycling conditions is of significant importance. This study systematically investigated the microstructural evolution and thermal properties of lunar regolith simulants under three distinct temperature cycling modes for 24 cycles: high-temperature cycling (25°C↔120°C), low-temperature cycling (25°C↔−120°C), and combined high-low temperature cycling (±120°C). The experimental results demonstrate that high-temperature cycling (HUST-2-H) primarily leads to an increase in both the quantity and dimensions of pores within the sample microstructure, increasing porosity by 11.7% (from 16.74% to 18.70%) and average pore size by 10.8% (from 4.28 μm to 4.74 μm). Conversely, low temperature cycling remarkably enhances the specific heat capacity by 15–20% across the measured temperature range. Notably, during combined high and low temperature cycling tests, the samples consistently showed a more pronounced response to low temperature effects. Furthermore, the synergistic interaction between alternating high and low temperature cycles effectively endowed the composite-cycled samples with significantly superior thermal insulation properties, yielding the lowest thermal conductivity (0.82 W/m·K at 25°C, 18.6% lower than control) despite significant elastic modulus reduction (20.8%), surpassing those achieved through single-temperature cycling processes. These findings not only elucidate the critical temperature cycling factors in lunar material formation and construction processes but also underscore the promising potential of lunar highland regolith as an effective thermal insulation material for lunar base infrastructure, with low-temperature effects dominating the response in combined cycling. These findings provide a preliminary evaluation of thermal cycling effects on sintered lunar regolith, offering practical insights for future lunar construction material selection and thermal design, and contribute to the growing body of knowledge on lunar regolith sintering and in-situ resource utilization.

1. Introduction

Since modern times, human exploration of extraterrestrial space has never ceased [1–3]. With the continuous advancement of deep space exploration missions, the Moon, as the closest celestial body to Earth and the first stop for deep space exploration, has shown great potential for resource development. Constrained by high transportation costs and the needs of future deeper space exploration, in-situ lunar base construction has become a key technology for sustainable exploration in recent years [4–6]. The Moon's surface experiences extreme temperature fluctuations ranging from approximately −173°C to 127°C, posing significant challenges to material integrity through mechanisms such as thermal fatigue

and differential thermal expansion [7–9].

Extensive research on lunar in-situ formation technologies has been conducted worldwide. Sintering has emerged as one of the most promising techniques for in-situ resource utilization in lunar construction [10–16]. For instance, Lin and Neves et al. successfully produced lunar soil concrete by mixing lunar soil or simulants with cement [6,11]; Fateri, Kim, and Ishikawa et al. used microwave sintering to achieve compressive strengths ranging from 12 to 120 MPa [17–19]; Microwave sintering was selected in this study due to its energy efficiency and potential for in-situ lunar application, as demonstrated in prior studies [20, 12–16,21]. Corrias et al. employed self-propagating high-temperature synthesis with reductants and additives under vacuum conditions to

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produce corresponding building materials [20]. Liu et al. incorporated NaOH solution into simulated lunar regolith and applied pressure during heating, achieving curing at low temperatures [22]. Additionally, Smithers et al. conducted full-scale construction experiments using a bagging confinement method [23]. Various sintering techniques using different heating methods have been explored [12], including hot-pressing [19,24], solar sintering [25], and laser selective melting [15]. However, each method has limitations. Microwave heating can overcome these issues due to its volumetric heating characteristics and deeper penetration depth [13,14,21].

However, most current studies focus on construction techniques and material preparation, with relatively little discussion on the performance of the formed components during service, particularly thermal insulation properties. Previous studies have primarily focused on lunar mare regolith simulants [13,26], while research on lunar highland regolith remains limited [20,22,23]. Given that numerous studies have fully demonstrated the feasibility of preparing bricks from lunar soil, it is of great significance to further explore the performance evolution of such bricks in the real extraterrestrial environment. The temperature conditions on the lunar surface differ greatly from those on Earth. The extreme fluctuations cause the temperature variation range to be approximately -173°C to 127°C , with a temperature difference of up to about 300°C . This poses a severe challenge to the performance of building materials. Therefore, studying the performance degradation of materials under intense thermal cycling, often exacerbated by thermal shock from rapid temperature changes, has become a key issue in the construction of lunar bases.

To simulate the effects of the extreme thermal environment on materials, this study selects a temperature range of -120°C to 120°C for thermal cycling tests. This range covers most typical temperature fluctuations on the lunar surface, helping to focus on the key thermal stresses that materials may encounter in actual service. By implementing three cycling modes—high-temperature cycling (room temperature to 120°C), low-temperature cycling (room temperature to -120°C), and combined high-low temperature cycling ($\pm 120^{\circ}\text{C}$)—this research systematically analyzes the influence of different temperature intervals on the material properties. This experimental design allows us to isolate the effects of the full thermal swing, the high-temperature segment, and the low-temperature segment separately, thereby identifying which temperature interval dominates material performance degradation under lunar surface conditions. In terrestrial research, there have been some explorations on the effects of thermal cycling on material properties. For example, Wang et al. evaluated the thermal stability of lightweight bulk metallic glasses after thermal cycling, suggesting their potential for application in near-Earth orbit environments [26]; Richard assessed the mechanical integrity of sulfur concrete specimens after thermal cycling, correlating strength changes with the coefficient of thermal expansion [7]; Xiong et al., based on a previously developed geopolymer gel model, revealed the stability mechanisms of geopolymer aluminosilicate structures under extreme temperature shocks from a molecular structure perspective [13]. Furthermore, studies focusing solely on high-temperature or low-temperature cycling are relatively common, such as Li et al.'s research on crack evolution in defective granite under high-temperature cycling [8], and Tomlinson et al.'s investigation of the early resistivity behavior of various concrete mixtures under low-temperature cycling [15]. However, research specifically aimed at extreme extraterrestrial service environments under thermal cycling remains scarce.

Current studies on lunar building materials under thermal cycling mostly focus on the effects of full high-low temperature cycles, with insufficient systematic analysis of the impacts of different temperature intervals (e.g., high-temperature segment, low-temperature segment). Therefore, this study designs three experimental conditions—high-low temperature cycling, high-temperature cycling, and low-temperature cycling—to comprehensively examine the effects of different thermal histories on the mechanical and thermal properties of materials and to

identify the critical influencing intervals. This aims to clarify whether high-temperature cycles, low-temperature cycles, or their coupled effects should be of greater concern in the production of lunar soil building materials and the construction of lunar bases, thereby providing guidance for subsequent material optimization and engineering practice. It is acknowledged that this study is preliminary in nature, focusing on macroscopic and microstructural trends rather than mechanistic depth, and aims to provide a foundational dataset for future in-depth investigations.

Throughout this paper, the following terminology is used consistently: 'high-temperature cycling' refers to cycling between room temperature and 120°C (samples designated HUST-2-H), 'low-temperature cycling' refers to cycling between room temperature and -120°C (samples designated HUST-2-L), and 'combined high-low temperature cycling' refers to cycling between -120°C and 120°C (samples designated HUST-2-C). The untreated control samples are designated HUST-2-B. These sample labels are defined in Table 1 and used consistently throughout figures, tables, and text (Table 2).

2. Experimental methods

2.1. Sample preparation and thermal cycling mode

Since the upcoming Chang'e-7 mission is scheduled to land in the lunar south polar region [3], this study selected sintered samples of HUST-2, a plagioclase lunar regolith simulant representing lunar highland and polar regions [20], as the experimental subject, with microwave sintering employed as the fabrication method. HUST-2 was developed based on HUST-1 [19] by mixing with anorthosite powder to represent lunar highland composition [20]. Compared to other lunar regolith simulants such as KLS-1 [13,26] and NAO-1 [22], HUST-2 specifically targets the highland regions relevant to the Chang'e-7 mission. The microwave sintering process was conducted as follows: the lunar regolith simulant was placed in a cylindrical graphite container, inserted into a microwave sintering furnace, and sintered at 1150°C for 2 h, followed by a 4-hour cooling period before the samples were retrieved. Three identical specimens were prepared for each thermal cycling condition to ensure reproducibility.

This experiment investigates the impact of thermal cycling on sample performance through three different temperature conditions: high-low temperature cycling, high-temperature cycling, and low-temperature cycling. As shown in Fig. 1, the temperature ranges for high-temperature cycling and low-temperature cycling are set from room temperature to 120°C and from room temperature to -120°C , respectively, while the high-low temperature cycling covers a range of $\pm 120^{\circ}\text{C}$. All cycling tests are set for 24 cycles.

The holding time of 5 h at extreme temperatures was selected based on thermal diffusion calculations and preliminary experiments to ensure temperature homogenization throughout the samples. The heating rate was approximately $1^{\circ}\text{C}/\text{min}$, and cooling was conducted passively (approximately 1 h from 120°C to room temperature, and similarly from -120°C to room temperature) to simulate the gradual temperature changes on the lunar surface rather than extreme thermal shock [9].

For high-low temperature cycling: samples were maintained at 120°C for 5 h, then kept at room temperature ($16 \pm 2^{\circ}\text{C}$) for 1 h, followed by 5 h at -120°C , and finally kept at room temperature for 1 h to complete one cycle. For high-temperature cycling only: samples were

Table 1
Sample thermal cycling information.

Thermal cycling mode	Sample label
High temperature	HUST-2-H
Low temperature	HUST-2-L
High and low temperature	HUST-2-C
No treatment	HUST-2-B

Table 2
Dimensions of samples before and after thermal cycling. (mean $\pm$ SD, $n = 3$).

Sample	Diameter before /mm	Diameter after /mm	Height before /mm	Height after /mm
HUST-2-B	10.02 $\pm$ 0.03	10.02 $\pm$ 0.03	20.02 $\pm$ 0.02	20.02 $\pm$ 0.02
HUST-2-H	10.01 $\pm$ 0.02	10.02 $\pm$ 0.02	20.01 $\pm$ 0.03	20.02 $\pm$ 0.03
HUST-2-L	10.03 $\pm$ 0.02	10.03 $\pm$ 0.03	20.00 $\pm$ 0.04	20.01 $\pm$ 0.03
HUST-2-C	10.02 $\pm$ 0.03	10.04 $\pm$ 0.03	20.01 $\pm$ 0.03	20.03 $\pm$ 0.05

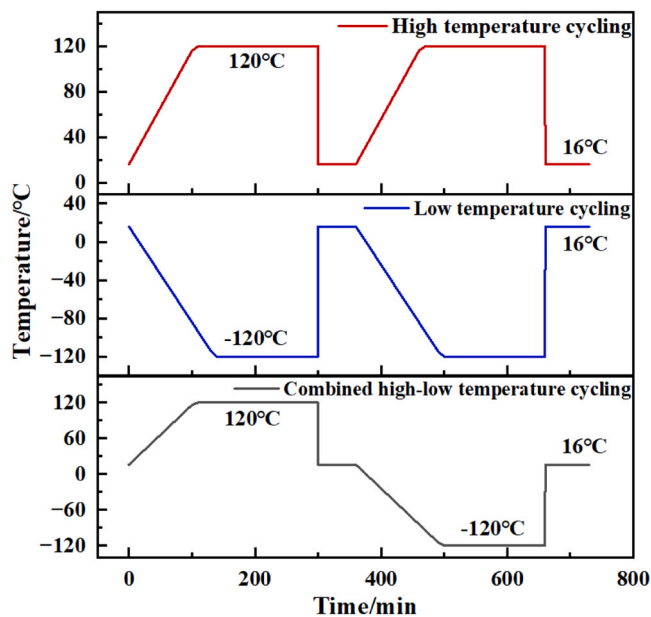


Fig. 1. Schematic diagram of the three thermal cycling modes. high-temperature cycling (16°C→120°C→16°C). low-temperature cycling (16°C→-120°C→16°C). combined high-low temperature cycling (16°C→120°C→16°C→-120°C→16°C).

maintained at 120°C for 5 h, then kept at room temperature for 1 h. For low-temperature cycling only: samples were maintained at -120°C for 5 h, then kept at room temperature for 1 h.

The samples of different materials and processing methods are numbered as shown in the following table:

2.2. Sample characterization

In this experiment, the universal material testing machine (Zwick Z200, Zwick/Roell Group Company of Germany) was used to test the mechanical properties, such as compressive strength and elastic modulus of the samples. Strain was measured using an extensometer attached to the specimen, with a loading rate of 0.5 mm/min. The elastic modulus was calculated from the stress and strain during the compression test. The thermal properties such as the thermal diffusivity and thermal conductivity of the samples were tested using a laser thermal conductivity meter (LFA467HT laser thermal conductivity meter, Netzsch Instruments, Germany). Three measurements were taken at each temperature point, and the average value was reported.

Phase and chemical composition analysis was carried out by X-ray diffraction (XRD; X 'Pert3 powder X-ray diffractometer (PANalytical B. V. of the Netherlands), with a scanning range of 20° to 70°, a step size of 0.02°, and a scanning speed of 5°/min. XRD was conducted to detect any potential phase transformations or crystallographic changes induced by thermal cycling. Porosity and pore distribution characteristics of the

samples were measured by mercury intrusion porosimetry (MIP). It should be noted that MIP measures only open porosity; closed pores are not detected by this technique [27,28]. The morphology and surface microstructure of the samples were studied using an environmental scanning electron microscope (SEM) (Nova Field Emission Scanning Electron Microscope, FEI Company, the Netherlands). Specimens for SEM were sectioned, polished, and gold-coated prior to observation to ensure representative surface imaging.

3. Results

3.1. Thermal cycling on quality and density

The samples before and after the temperature cycling treatment are shown in Fig. 2. No visible changes were observed on the sample surface, indicating that the temperature cycling treatment method adopted in this experiment does not have a clearly visible impact on the samples.

After the sample undergoes temperature cycling treatment, the variation curves of its mass and density with the number of cycles are shown in Fig. 3, where the density is calculated based on the mass and size. It should also be noted that the mass and density of the control samples (HUST-2-B) were 3.695 g and 3.136 g/cm³, respectively. As shown in Fig. 3(a), the masses of the thermally cycled samples (HUST-2-H, HUST-2-L, and HUST-2-C) all presented a slightly decreasing trend with fluctuations. Among them, the self-mass difference of sample HUST-2-C before and after the high and low temperature cycling treatment was approximately 0.004 g. The mass difference between the samples HUST-2-H and HUST-2-L before and after the high temperature and low temperature cycling treatment varies from 0.001 g to 0.003 g. The mass loss of all samples is very small. Moreover, since the vernier caliper used for measurement can only measure to three decimal places, the curve fluctuation can be regarded as a measurement error to a certain extent. Error bars representing standard deviation from three replicate measurements are included in Fig. 3, though variations were minimal.

As shown in Fig. 3(b), the sample HUST-2-C, which underwent combined high-low temperature cycling, exhibited a noticeable downward trend in density. Sample HUST-2-C exhibited a statistically significant downward trend in density ($p < 0.05$), with a total decrease of 0.0131 g/cm³ after 24 cycles. Although the absolute change is small, representing only 0.56% of the initial density, the consistent downward trend suggests a measurable effect of combined cycling on density. Following the cyclic treatment at high and low temperatures, the densities of the samples HUST-2-H and HUST-2-L displayed an overall fluctuating trend, without any significant increase or decrease. This may be because the high temperature or low temperature treatments had

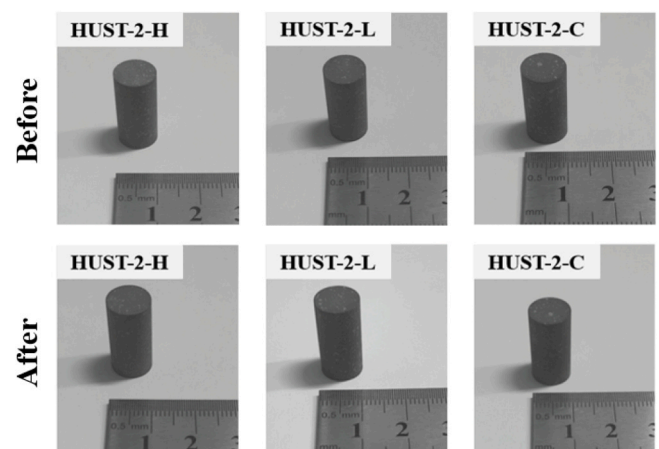


Fig. 2. Samples before and after temperature cycling.

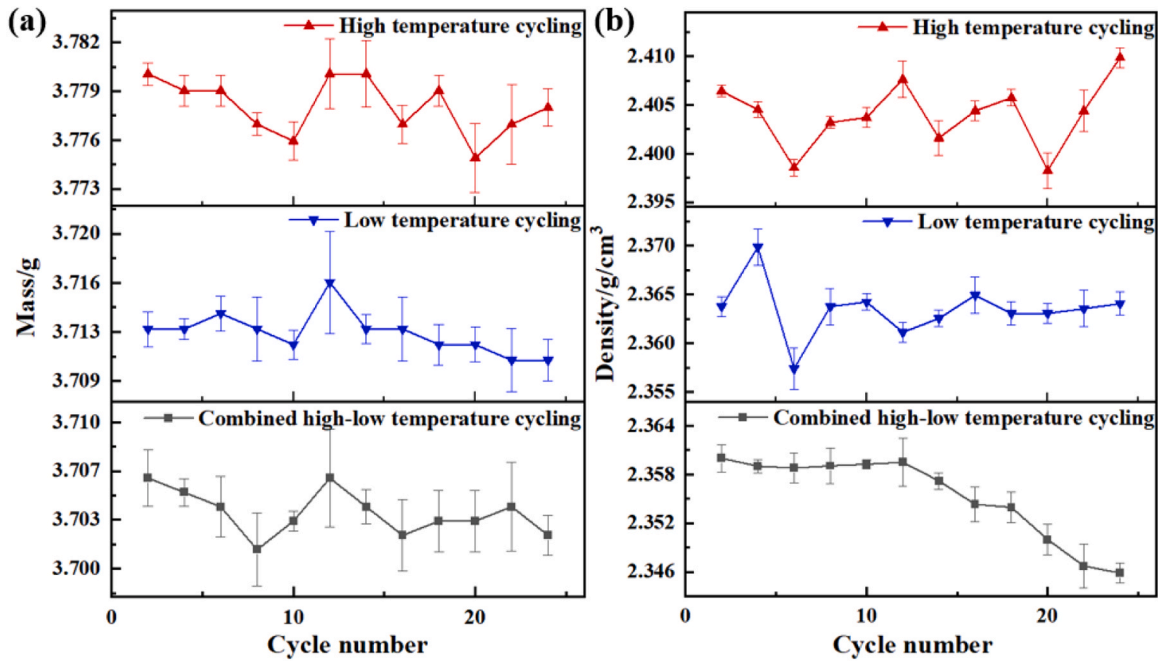


Fig. 3. (a) Mass change and (b) density change of the sample with the number of cycles.

minimal impact on the physical dimensions, such as diameter and height, resulting in negligible changes in the calculated density. The cycling temperatures may have caused a slight loss of sample components, leading to very minor increases in diameter and height. The small degree of mass loss and density change suggests that the samples can maintain basic stability under substantial temperature cycling. This indicates that buildings constructed from lunar soil sintered bricks could remain stable despite the challenges posed by extreme temperature fluctuations in extraterrestrial environments, which is favorable for the advancement of lunar construction.

3.2. Thermal cycling effects on microstructure

Fig. 4 shows the XRD diffraction patterns of the experimental samples. The main phase in all samples is plagioclase feldspar, consistent with the composition of HUST-2 simulant and other lunar highland

simulants [19,20,22,23]. After thermal cycling, all samples retain the same diffraction peak positions as the control, indicating that no phase transformation occurred during any of the cycling treatments. Minor variations in peak intensity were observed in some angular ranges (e.g., 23–24° and 27–28°), but these variations were within the range of normal experimental variability and did not show systematic trends across replicate measurements. Therefore, we conclude that thermal cycling did not induce detectable changes in phase composition or crystallographic orientation under the conditions tested. This stability is consistent with observations in other sintered lunar regolith studies [13, 29,25].

Fig. 5 shows the scanning electron microscopy (SEM) images of all samples. In the figure, images (a)–(d) were taken at a magnification of $\times 200$, while images (e)–(h) were taken at $\times 5000$. The high-magnification areas correspond to the relatively flat regions observed under low magnification.

As shown in Fig. 5(a) to (d), the control sample HUST-2-B exhibits large interconnected concave regions, resulting in an uneven surface morphology. The diameter of these surface depressions is approximately 15 μm . In comparison, the surface flatness of samples HUST-2-H, HUST-2-L, and HUST-2-C improved to varying degrees, mainly characterized by a reduction in large concave areas and the appearance of deeper, smaller-diameter holes with well-defined edges. This indicates that thermal cycling has a positive effect on the surface flatness of the samples. However, no significant differences were observed among the three thermal cycling methods. Fig. 5(e) to (h) reveal that the surface of sample HUST-2-B displays a planar-like morphology with clear, interconnected edges. The edges of sample HUST-2-H show slight rounding suggestive of surface softening, while those of sample HUST-2-L appear relatively defined and angular. Sample HUST-2-C exhibits a higher degree of surface flatness. Compared to HUST-2-B, the slightly passivated edges of HUST-2-H may be attributed to minor melting on the sample surface caused by high-temperature treatment. Sample HUST-2-L did not show notable changes in its microscopic morphology. Sample HUST-2-C likely represents a small region that underwent slight melting during the high-low temperature treatment, followed by rapid cooling and solidification, leading to gradual surface fusion and enhanced flatness.

As shown in Fig. 6, the main pore diameters of samples HUST-2-B, HUST-2-L, and HUST-2-C are all approximately 4 μm , while that of

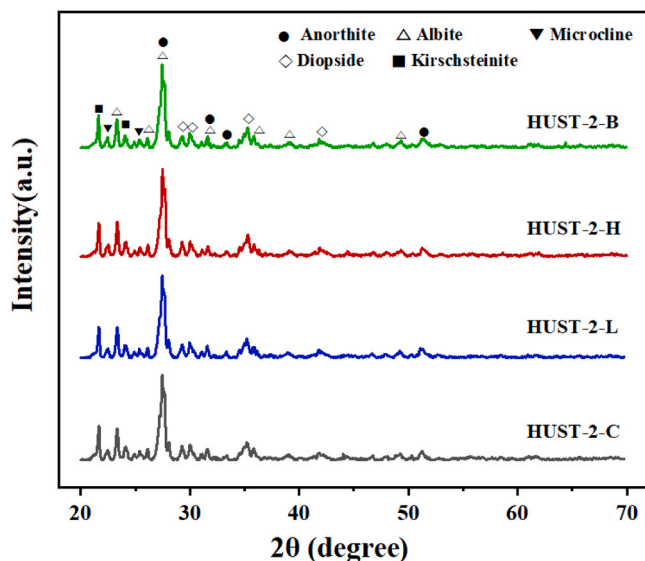


Fig. 4. XRD diffraction pattern of the samples.

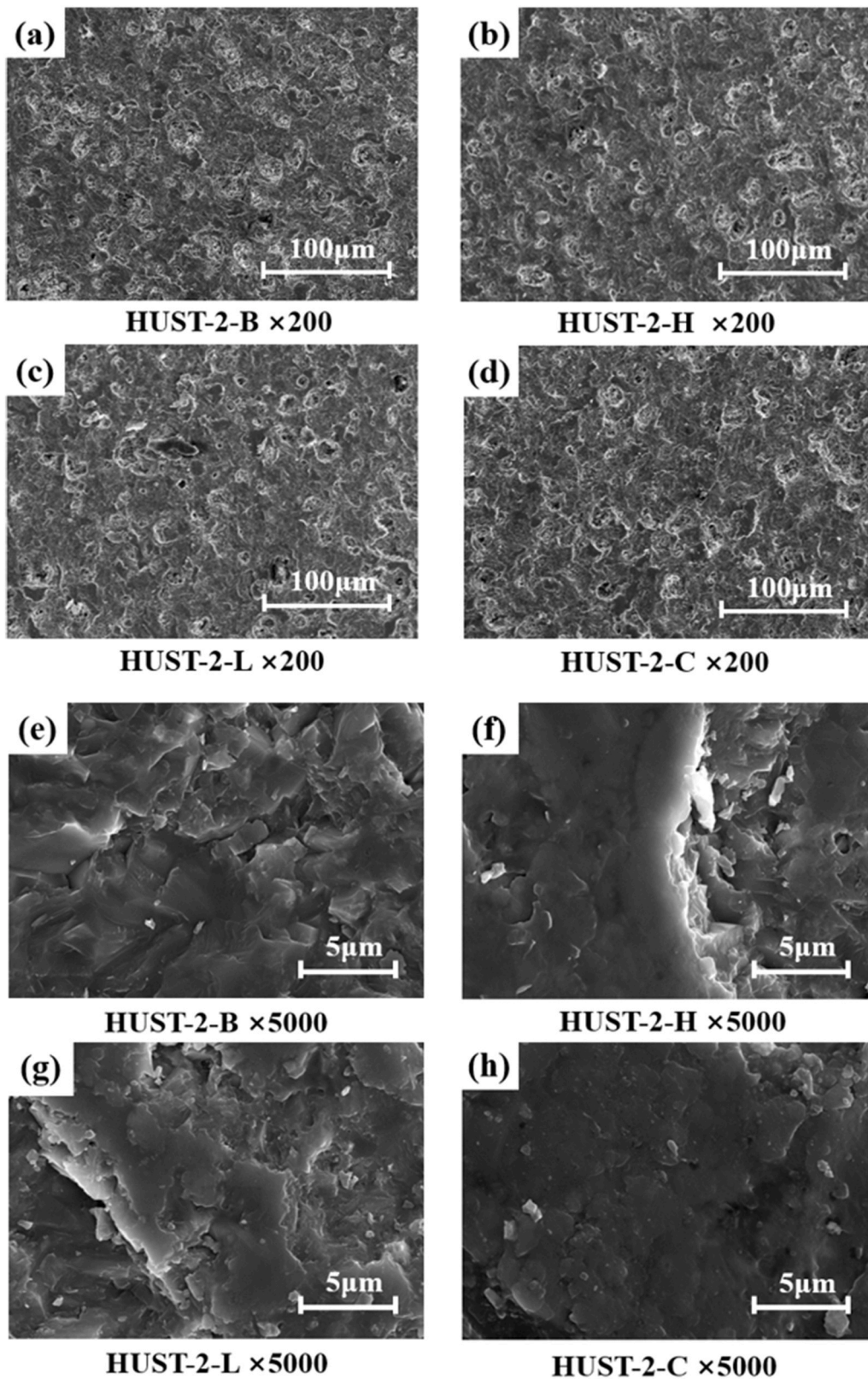


Fig. 5. Scanning electron microscope observation of the samples.

sample HUST-2-H is slightly larger, at nearly 5 μm . Moreover, in terms of pore size distribution, sample HUST-2-H exhibits a significantly higher number and proportion of pores compared to the other samples. A trend is observed at a maximum diameter of 800 μm when comparing samples HUST-2-L and HUST-2-C, as well as HUST-2-B and HUST-2-H. However,

no pores approaching 1 mm were detected during compression tests. This suggests that the apparent trend in the curve may result from interconnected smaller pores or connections formed under increased pressure during mercury intrusion, giving the impression of larger pores. Fig. 5 shows that surface depressions observed via SEM can reach up to

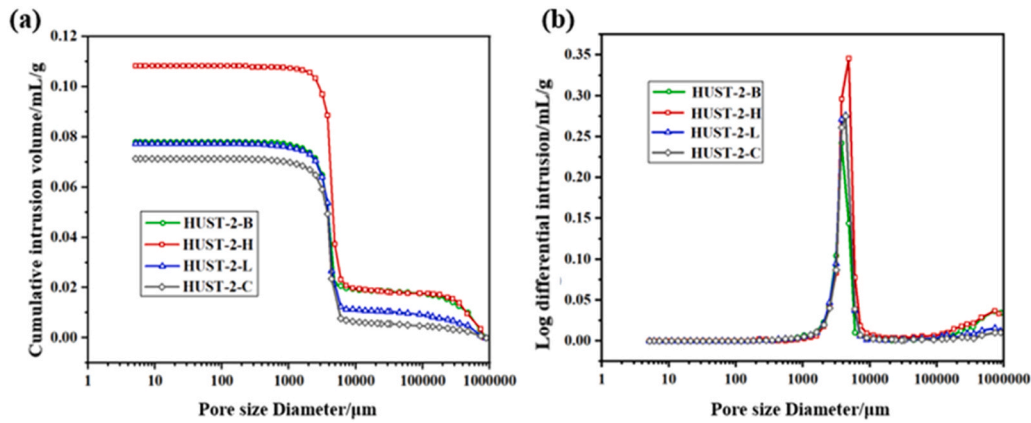


Fig. 6. (a) Cumulative intrusion curves. and (b) logarithmic differential intrusion curves. of the sample.

15 μm in diameter, whereas the porosity test reveals almost no trend in this size range. This discrepancy indicates that the internal pore morphology assessed by MIP differs from the surface topography observed by SEM, likely because SEM captures surface features that are not fully representative of the interconnected pore network within the bulk material. Therefore, it is inferred that the vast majority of pores in the samples are closed, meaning that internal pore structures exert a greater influence on overall performance.

Porosity values are presented as mean $\pm$ standard deviation based on triplicate measurements.

In Fig. 6(a), the cumulative intrusion volumes of samples HUST-2-L and HUST-2-B are nearly identical, while that of HUST-2-C is only slightly lower. Notably, the curve for HUST-2-B shows a smaller drop in the main pore diameter region. This is further illustrated in Fig. 6(b), where the peak value of the HUST-2-B curve is the lowest, indicating that its corresponding pore diameter is also the smallest. In terms of average pore diameter and porosity, samples HUST-2-L and HUST-2-C show a significant reduction in average pore size compared to the control group HUST-2-B, though the changes in porosity are less pronounced. Considering the trend exhibited by HUST-2-B in the large-diameter region, it appears that the average pore diameters of both HUST-2-B and HUST-2-H are influenced by this trend and thus appear larger. Although the average pore size of HUST-2-B should be similar to that of HUST-2-L and HUST-2-C, a notable increase in both average pore size and porosity is clearly observed in sample HUST-2-H.

The results indicate that for HUST-2 samples, low-temperature cycling may lead to a slight increase in the number of tiny internal pores. In contrast, high-temperature cycling can increase both the number and diameter of tiny pores, with the effects of high-temperature cycling being more pronounced. However, when analyzing data such as the proportion of primary pores and overall porosity, along with visual comparisons in the figures, it becomes evident that the pore characteristics of samples subjected to combined high- and low-temperature cycling closely resemble those treated with low-temperature cycling alone. This suggests that for bricks formed from lunar highland regolith, the low-temperature cycling exposure plays a dominant role in pore formation. In some respects, the influence of the low-temperature period may even overshadow that of the high-temperature period.

3.3. Thermal cycling effects on mechanical properties

The compressive strength and elastic modulus of the specimens were evaluated through compression tests conducted using a universal testing machine. Fig. 7 shows the stress-strain curves obtained from the compression tests. The initial non-linear portion of the curves up to approximately 500 microstrain is common in brittle sintered materials and may be attributed to seating errors or initial closure of pre-existing microcracks. The compressive strength was determined as the maximum

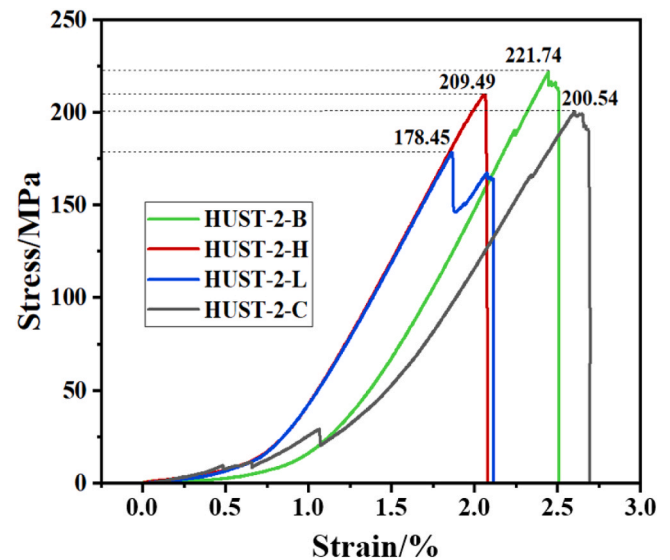


Fig. 7. Compression stress-strain diagram of the sample.

stress recorded during the compression process, while the elastic modulus was calculated from the slope of the linear elastic region in the stress-strain curve. As shown in Fig. 7, all specimens except HUST-2-H exhibited a noticeable stress drop before failure. This trend was particularly evident in samples HUST-2-L and HUST-2-C. It can be inferred that internal defects in these specimens led to early damage in localized areas during compression, resulting in an overall stress reduction. Had these defects been absent, the strength of HUST-2-L and HUST-2-C could potentially have been higher.

Although subjected to three different temperature cycling conditions, the compressive strengths of HUST-2-H, HUST-2-L, and HUST-2-C remained similar. However, compared to the control sample HUST-2-B, all thermally cycled specimens showed a significant reduction in compressive strength. Interestingly, although sample HUST-2-H exhibited higher porosity and larger internal pore sizes than HUST-2-L and HUST-2-C, its compressive strength did not decrease as might be expected. This phenomenon, where increased porosity does not proportionally reduce strength, has been observed in other sintered lunar regolith studies and is attributed to pore morphology and distribution [12,29,30].

The curves depicting the elastic modulus and its degradation rate for the samples are presented in Fig. 8. The degradation rate was calculated as the difference between the elastic modulus of the control group and that of the experimental group, divided by the modulus of the control

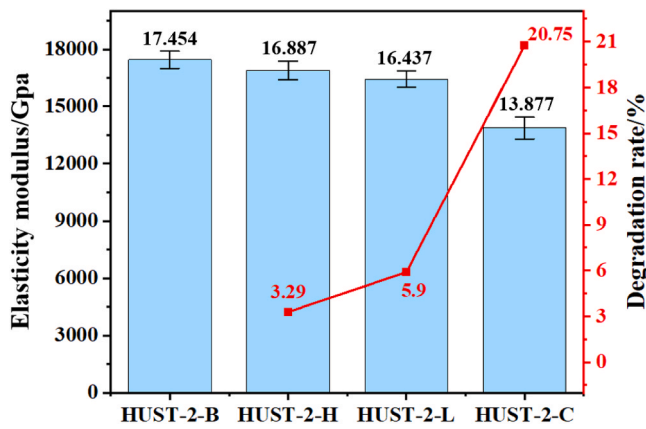


Fig. 8. Elastic modulus and degradation rate of the sample.

group. As shown in Fig. 8, the elastic modulus of samples HUST-2-H and HUST-2-L decreased to some extent compared to HUST-2-B, with HUST-2-L exhibiting a slightly greater reduction. In contrast, the decline for HUST-2-C was much more substantial, reaching 20.75%. The smaller temperature variation range in the high-temperature cycle may account for the relatively smaller decrease in elastic modulus observed in HUST-2-H compared to HUST-2-L. However, given the pronounced degradation in HUST-2-C, it is reasonable to conclude that the temperature variation range in individual high- or low-temperature cycles has limited influence on the elastic modulus. Instead, the combined effect of coupled high- and low-temperature cycles with an excessive temperature range appears to significantly reduce the elastic modulus. Fan et al. investigated crack formation in granite under extreme temperature variations and suggested that, within a certain range, thermal shock damage increases with the temperature range [8]. They hypothesized that as the temperature range of the cycling mode increases beyond a threshold, the deterioration of the elastic modulus transitions from an approximately linear trend to one resembling exponential degradation. This analogy suggests that thermal fatigue, driven by the large temperature swing in the combined cycling, may be a key mechanism for the observed modulus reduction in HUST-2-C.

3.4. Thermal cycling effects on thermal performance

As illustrated in Fig. 9, the thermal diffusion coefficients of all

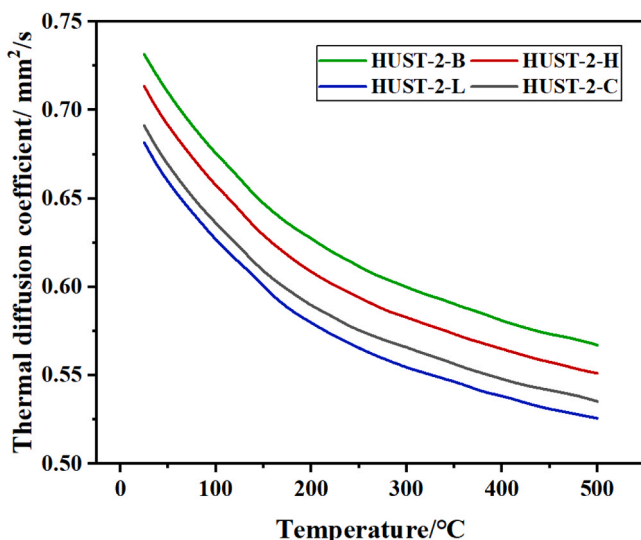


Fig. 9. Thermal diffusion coefficient of the sample.

samples decrease with increasing test temperature, exhibiting a trend resembling an inverse proportionality function. The sample labeled HUST-2-B, serving as the control group, shows the highest thermal diffusivity among the curves. Following temperature cycling treatment, all thermally cycled samples exhibit varying degrees of reduction in thermal diffusivity. Specifically, sample HUST-2-H demonstrates the smallest decrease, while HUST-2-L shows the largest decline. The curve of HUST-2-C lies between these two, slightly closer to that of the low-temperature cycled sample HUST-2-L, with the proximity becoming more pronounced below 150°C—a temperature relevant to the lunar surface environment.

The specific heat capacity curve of the HUST-2 sample, presented in Fig. 10, displays fluctuations in contrast to the relatively smooth thermal diffusivity profile. Despite these variations, a clear overall trend of increasing specific heat capacity with rising test temperature is observed. This trend becomes particularly evident after the temperature exceeds 500°C. The specific heat capacity values obtained are consistent with those reported for other lunar regolith simulants [31–33]. Among the samples, HUST-2-H shows a slight decrease in specific heat capacity, HUST-2-C a moderate increase, and HUST-2-L a marked increase, suggesting that low-temperature cycling enhances the sample's thermal stability at elevated temperatures. Based on the observed trends, it can be inferred that the specific heat capacity of all samples would continue to decrease in a low-temperature environment. Furthermore, Figs. 8 and 9 show that the curve of HUST-2-C lies between those of HUST-2-H and HUST-2-L, indicating that both high- and low-temperature cycling influence the thermal properties of the material. When combined with porosity and compression test results, it becomes clear that the effects of low- and high-temperature cycling are interactive, each distinctly affecting different material characteristics. While the observed trends are consistent across replicates, further statistical validation with larger sample sizes is recommended for definitive conclusions regarding the effect of thermal cycling on specific heat capacity.

4. Discussion

In the compression tests of the HUST-2 samples, HUST-2-H is the only specimen that does not exhibit a stress drop during loading. The mechanical behavior of HUST-2-H presents an interesting case. SEM images (Fig. 5f) reveal that this sample has more pronounced surface features and defects compared to others, which might be expected to reduce strength. However, its compressive strength is not significantly lower than other samples. This apparent paradox may be explained by considering that in sintered materials, compressive failure is often

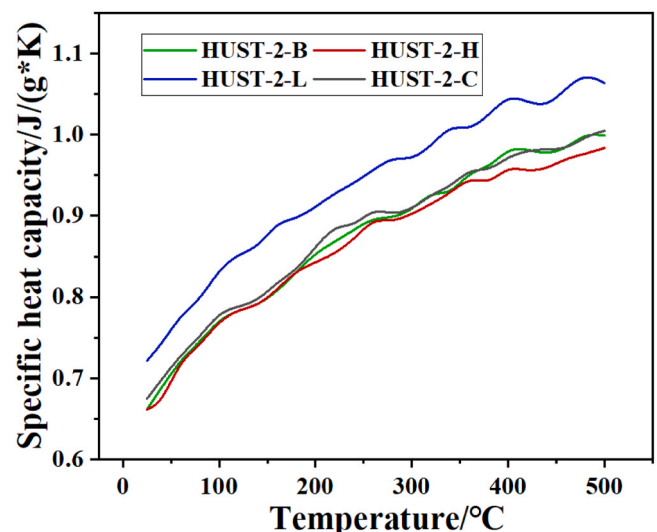


Fig. 10. Specific heat capacity of the sample.

governed by the internal pore structure rather than surface morphology [27,34]. While HUST-2-H has higher total porosity (18.70%), MIP data suggest a relatively narrow pore size distribution (Fig. 6b), indicating uniform internal pore sizes. If these pores are predominantly closed and evenly distributed, they may not create the stress concentration sites that lead to premature failure. In contrast, samples with lower total porosity but more heterogeneous pore distributions (such as HUST-2-L and HUST-2-C, which showed stress drops) may contain localized defect clusters that trigger failure despite lower overall porosity. Overall, the influence of the three temperature cycling modes on compressive strength and elastic modulus via pore structure modification appears limited. Thermal fatigue in sintered ceramics typically involves progressive microcracking due to cyclic thermal stress. In this study, the combined high-low cycling likely induced more severe stress gradients, leading to greater modulus degradation in HUST-2-C, as supported by prior work on granite [8].

$$\lambda = \rho \alpha c \quad (1)$$

c is specific heat capacity, $J/(g \cdot K)$;

λ is thermal conductivity, $W/(m \cdot K)$

α is the thermal diffusivity coefficient, mm^2/s ;

ρ is density, g/mm^3

Thermal conductivity can be derived from thermal diffusivity and specific heat capacity, enabling observation of its temperature dependence. As shown in Fig. 11, with increasing temperature, thermal diffusivity decreases, whereas thermal conductivity tends to increase. Despite some point-to-point fluctuations arising from measurement variability, a clear overall trend of increasing thermal conductivity with temperature is observed for all samples. Given the study's focus on thermal insulation and heat retention—particularly relevant to building applications—special attention is given to thermal conductivity. Among the samples, HUST-2-C exhibits the lowest thermal conductivity, indicating superior heat preservation performance. This suggests that under high- and low-temperature cycling, the thermal insulation properties of sintered HUST-2 samples made from lunar regolith simulant improve, implying that highland lunar soil may contribute to maintaining temperature stability within structures in a lunar base (Fig. 12).

Summarizing the above results, the effects of the three thermal cycling modes on HUST-2 samples can be compared as follows: high-temperature cycling primarily affects pore structure, significantly increasing average pore size (+10.8%) and porosity (+11.7%); low-temperature cycling mainly influences thermal properties, markedly enhancing specific heat capacity (+15–20%) with minimal impact on

pore structure; combined cycling, while maintaining porosity essentially unchanged, significantly reduces elastic modulus (−20.8%) and produces the best thermal insulation performance (lowest thermal conductivity). This suggests that high temperature dominates pore evolution, low temperature dominates thermal property changes, while their coupling primarily affects mechanical properties with mutually offsetting effects on microstructure. This comparison clearly demonstrates the distinct contributions of each temperature interval and the synergistic effects of combined cycling.

Regarding the thermal conductivity and specific heat capacity trends, when the density of HUST-2-C is similar to that of HUST-2-L, the notably higher specific heat capacity of HUST-2-L leads to a clear difference in thermal conductivity. Combined with Fig. 11 and pore structure analysis, it is inferred that during high-temperature cycling, thermal expansion may induce microcracks around pores, leading to pore growth and interconnection. The porosity data in Fig. 6 and Table 3 support this mechanism: the increased pore size ($4.74 \mu m$ vs. $4.28 \mu m$ in control) and porosity (18.70% vs. 16.74% in control) in HUST-2-H are consistent with thermal expansion-induced microcracking around pores during high-temperature exposure. The thermal stress generated during heating ($\sigma \approx E \cdot \alpha \cdot \Delta T$, estimated at ~ 9.6 MPa) may be sufficient to initiate microcracking at stress concentration points around existing pores. In contrast, during low-temperature cycling, shrinkage stresses may remain below the fracture threshold, leaving residual stresses that influence specific heat instead. The marked increase in specific heat capacity observed in HUST-2-L (Fig. 10) aligns with this interpretation. The mechanism by which residual compressive stresses increase specific heat capacity can be understood through lattice strain effects. Residual stresses induce slight changes in lattice parameters, altering the anharmonicity of interatomic potentials and thus affecting the phonon spectrum. This lattice strain increases the thermal energy required per unit temperature rise, manifesting as increased specific heat capacity. This effect is particularly pronounced when stresses remain below the fracture threshold, consistent with the observation that HUST-2-L shows little porosity change but significantly enhanced specific heat. In combined high-low temperature cycling, cracks formed during the high-temperature phase may be mitigated or healed in the subsequent low-temperature phase, resulting in no net increase in porosity. However, whether cracks formed during high-temperature exposure are partially mitigated during low-temperature phases, or whether the stress state in combined cycling simply does not favor permanent pore enlargement, remains a question for future investigation. At present, this is offered as a hypothesis requiring further validation rather than a demonstrated mechanism.

According to Formula (1) and the thermal diffusivity/thermal conductivity curve shown in Fig. 11, it can be seen that although HUST-2-H has a higher density and thermal diffusivity than HUST-2-L, its thermal conductivity is actually lower - this might be due to the influence of pores. Generally speaking, an increase in porosity will lead to a decrease in thermal conductivity [24]. As the sample was sintered under low pressure, the closed pores inside it contain very little air and are in a sub-atmospheric pressure state [25]. Due to the poor thermal conductivity of air, the effective thermal conductivity coefficient within the pores is significantly lower than that of the solid matrix. These pores thus form thermal barriers, hindering heat transfer. Thus, the high porosity of HUST-2-H exhibits a lower thermal conductivity, while the HUST-2-C with the lowest porosity and average pore size shows the lowest thermal conductivity, which is consistent with this model. These findings to some extent confirm the correlation between the changes in pore structure caused by thermal cycling and the fluctuations in thermal conductivity.

As test temperature rises, thermal diffusivity decreases, while thermal conductivity increases. Since this study focuses on thermal insulation and retention—particularly in the context of building envelopes—thermal conductivity is a key parameter. Among all samples, HUST-2-C displays the lowest thermal conductivity and a clear

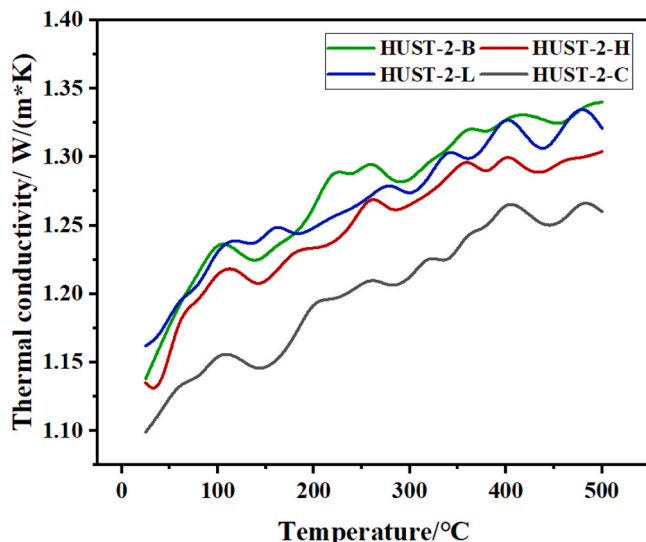


Fig. 11. Thermal conductivity of the sample.

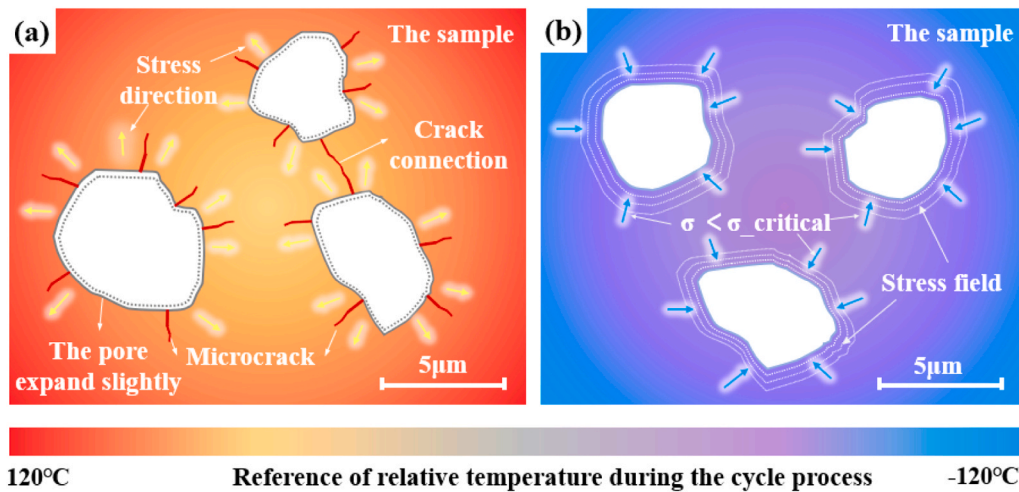


Fig. 12. Speculative schematic diagram of. (a) The cracks produced in the samples during high-temperature cycling and. (b) The pores of the samples during low-temperature cycling remain almost unchanged.

Table 3

Sample pore information(mean ± SD, n = 3).

Sample label	Porosity/%	Average pore size/μm
HUST-2-B	16.74 ± 0.21	4.28 ± 0.15
HUST-2-H	18.70 ± 0.28	4.74 ± 0.18
HUST-2-L	16.56 ± 0.19	3.76 ± 0.12
HUST-2-C	16.05 ± 0.22	3.56 ± 0.14

reduction compared to HUST-2-B, suggesting it has the best heat preservation performance. Thus, under repeated high-low temperature cycles, the thermal insulation capability of the sintered HUST-2 lunar simulant samples appears enhanced. This further supports the idea that highland lunar regolith could help stabilize internal temperatures in lunar base structures.

From an engineering perspective, the findings of this study have practical implications for lunar base construction. Based on the thermal conductivity of HUST-2-C (approximately 0.8 W/m·K) and the extreme temperature swing on the lunar surface (approximately 300°C), preliminary calculations suggest that maintaining a 20°C temperature differential inside a habitat would require an insulation layer thickness of approximately 0.5–1.0 m, depending on internal heat sources and external conditions [6,11]. This thickness is engineering-feasible and could be achieved through multi-layer brick construction. Regarding long-term durability, while 24 cycles have revealed performance trends, the lunar surface experiences approximately 12 day-night cycles per year [9], meaning a structure designed for 10 years would endure about 120 cycles. The most significant elastic modulus degradation observed in this study (20.8% in combined cycling) suggests that long-term service may face stiffness reduction issues. Future studies should extend cycling numbers to 100 + cycles to evaluate long-term performance evolution, as recommended in recent ISRU assessments [6,10].

This study, while providing valuable preliminary insights, has several limitations that should be acknowledged. First, the sample size (three specimens per condition) limits statistical power for definitive conclusions. Second, the thermal cycling profile employed (24 cycles with fixed 5-hour holding times) simplifies the complex thermal environment on the lunar surface, which involves diurnal cycles, partial shading, and more rapid temperature changes [9]. Third, the experiments were conducted at ambient pressure rather than under lunar vacuum conditions, which may affect heat transfer and microstructural evolution [30]. Fourth, no in-situ monitoring was performed during thermal cycling, limiting our ability to observe dynamic changes. Fifth, microstructural observations were limited to sample surfaces. Polished

cross-sectional imaging would provide valuable information about internal pore structure and closed porosity, and should be included in future investigations. Sixth, true density measurements would be required to quantify total porosity and closed pore fraction [27].

Future work should address these limitations through: (1) larger sample sizes for statistical validation; (2) extended cycling numbers (e.g., 100 + cycles) to investigate long-term degradation; (3) variable ramp rates to simulate more realistic thermal shock conditions; (4) testing under vacuum to better approximate lunar conditions [30]; (5) in-situ microstructural monitoring during thermal cycling using techniques such as high-temperature XRD or environmental SEM; (6) direct measurement of residual stresses (e.g., via Raman spectroscopy or XRD peak broadening) to validate the proposed mechanisms; and (7) polished cross-sectional imaging to characterize internal pore structure. Such investigations would provide a more comprehensive understanding of material behavior under lunar thermal conditions and contribute to the development of reliable construction materials for future lunar bases [10–12].

5. Conclusion

Temperature cycling treatments significantly affect the microstructure and thermodynamic properties of microwave-sintered simulated highland lunar soil samples, with variations in impact depending on the cycling method. High-temperature cycling (HUST-2-H) increases porosity by 11.7% and pore size by 10.8% through thermal expansion-induced microcracking. Low-temperature cycling (HUST-2-L) enhances specific heat capacity by 6–10% via residual shrinkage stresses without significant microstructural damage. Combined cycling (HUST-2-C) produces the best thermal insulation performance (the thermal conductivity is approximately 7% below control) despite 20.8% elastic modulus degradation, with low-temperature effects dominating the response. The samples experience the combined effects of high and low temperatures throughout the temperature cycling process. It appears that the impact of low-temperature cycling is more pronounced. Additionally, the interaction between high and low temperature cycles results in in HUST-2-C enhanced thermal insulation properties, suggesting that polar and highland lunar soils could effectively preserve heat and provide good thermal insulation in environments with significant temperature fluctuations. These findings suggest that lunar highland regolith bricks could provide effective thermal insulation in lunar bases, though long-term performance under extended cycling requires further investigation. The results contribute to the growing understanding of lunar regolith sintering and in-situ resource utilization [10,11,20,12],

and provide a foundation for future material optimization for lunar construction. This characteristic will be beneficial for future lunar base construction. These results provide a foundational understanding of thermal cycling effects on sintered lunar regolith, highlighting the need for further investigation into thermal fatigue mechanisms and long-term performance under more realistic lunar conditions.

CRedit authorship contribution statement

Chu Wang: Formal analysis, Data curation. **Bo Pan:** Formal analysis, Data curation. **Linxuan Jiang:** Writing – original draft, Data curation. **Fen Dang:** Writing – original draft, Formal analysis. **Yan Zhou:** Writing – review & editing, Supervision, Funding acquisition. **Cheng Zhou:** Writing – review & editing, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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